

An Enhanced EfficientNetB0 Framework for Pneumonia Detection from Chest X-ray Imagery

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Abstract—Pneumonia is a severe respiratory illness that continues to be a significant cause of morbidity and mortality on a global scale, with a particular emphasis on minors and the elderly. Manually interpreting chest X-ray pictures is laborious and requires radiological knowledge, but an early and correct diagnosis is crucial for successful therapy. In this work, we propose an improved deep learning framework for automated pneumonia identification from chest X-ray pictures, which is based on EfficientNetB0. To develop a simplified approach and reduce over-fitting, the proposed EfficientNetB0 employs a pre-trained EfficientNetB0 design with ImageNet weights as an attribute extractor, followed by batch normalization, fully connected layers with L1/L2 regularization, and dropout. Transfer learning is used to capitalize on rich visual representations gained from large-scale image collections. We use public chest X-ray datasets with normal and pneumonia classes to train and test the model. As shown experimentally, the suggested framework is effective, with 96.80% test accuracy, 96.43% validation accuracy, and 99.11% training accuracy. In addition, the model achieves very good results for the Pneumonia class, with a recall of 0.97, an F1-score of 0.98, and an overall weighted F1-score of 0.97. There were few erroneous predictions, and the classification report shows a high degree of discrimination between healthy subjects and those with pneumonia.

Keywords—Images, Efficient NetB0, Convolutional Neural Networks (CNN) and Bacteria-Pneumococci etc.

I. INTRODUCTION

Pneumonia is a discriminating respiratory syndrome that is caused by infectivity and inflammation of the lungs. The disease may be attributed to several organisms such as bacteria, viruses, fungi and parasites [1]. Infection causes growth of the air sacs as well as sometimes fluid gather, impairing the ability of the air sacs to exchange oxygen with the blood. Patients usually develop symptoms like fever, chills, difficulty in breathing, chest pain, fatigue, and coughing that will persist [20]. Streptococcus pneumonia and Haemophilus influenza are ordinary microbes pathogens linked to pneumonia cases and antibiotic treatment is generally recommended. Influenza viruses and SARS-2-2019 could all reason viral pneumonia [11]. Depending on the severity of the disease, antiviral medications, supportive therapy and close clinical observation are used for management, as antibiotics are not effective in treating viral infections [12].

Efficient-Net has emerged as a powerful and computationally efficient model due to its compound scaling method, which uniformly scales network depth, width, and resolution [14]. Efficient-Net achieves high classification accuracy while maintaining lower computational complexity compared to traditional CNN models. Respiratory problem detection is impacting people in several significant ways [13]. Pneumonia's a severe lung infection that could be serious, particularly for patients with compromised immune systems, the elderly, and infants. Efficient-Net and other advanced deep

learning models are making It facilitates the diagnosis of pneumonia in chest X-ray quickly and accurately [14].

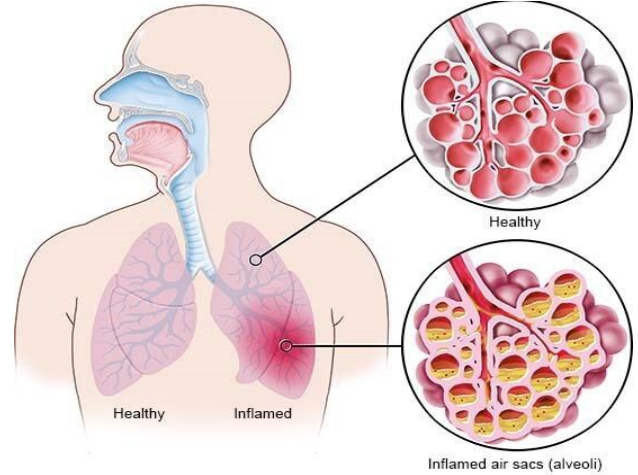


Fig. 1. Pneumonia [21].

A. Causes and Typical Symptoms of Pneumonia Include

Most cases of pneumonia are caused by bacteria shown in Fig. 1. In most cases, the site of infection dictates the kind of bacteria: Pneumococci are the most common bacteria that might cause pneumonia while you're at home. Pneumonia at a healthcare facility is often due to other, more difficult-to-treat microorganisms.

- High Temperature,
- Lethargy,
- Coughing up Mucus,
- Difficulty Breathing

B. Importance of Chest X-Ray

A non-surgical, readily available, and cost-effective diagnostic tool is chest X-ray imaging. When it comes to identifying lung diseases, such as pneumonia, it is vital [16]. Medical professionals can investigate the lung fields for anomalies using chest X-rays. Particularly concerning in the context of lung disease, these anomalies may manifest as regions of elevated opacity, suggesting the presence of fluid buildup and inflammation [17].

Serious complications, including infection, collapse of the circulatory heart, or irregular heartbeat, are possible [18] [19]. In cases when the patient is either under two years old or over 65 years old, has a damaged immune system, requires artificial breathing, has contracted pneumonia while hospitalized, has concomitant medical issues such as renal or cardiac sickness, or is in need of mechanical ventilation [23].

The preceding section provides an overview of the research work. The next part will go over the review of deep learning and machine learning for pneumonia diagnosis. Section IV explores the outcome and substantiation, whereas

Section III covers the recommended technique. Section V discusses the method's conclusion and future scope.

II. REVIEW OF LITERATURE

In the present section, we will examine a literature review that discusses the use of innovative methods and machine learning for pneumonia identification. Table I summarizes the research, its methods, dataset, important results, and drawbacks.

Opeyemi Usman et al. (2025): explored an automatic pneumonia detection system using the transfer learning method with Efficient-Net architecture. Pneumonia is still one of the leading causes of respiratory death worldwide, especially in areas where there are fewer access to expert radiological services. The researchers used a publicly available database of CXR images from Kaggle (Pneumonia) comprising 5,856 images. A series of data splitting strategies were tested to find the best training split, such as 80:10:10, 70:15:15, 60:20:20, or 75:15:10. The image pre-processing consisted of normalization and augmentation, and the model was trained for 25 epochs [1].

Yashvi Chandola et al. (2025): This study used an improved version of the Kaggle dataset, which consisted of ~18,200 chest X-ray images. The performances of three lightweight CNN networks ShuffleNet, NASNet-Mobile, and Efficient-NetB0 were compared through transfer learning. The classification accuracy achieved for each was 90%, 88%, and 89% respectively. The researchers then took the deep feature representations from each network and combined them with classifiers SVM and XG-Boost to further enhance the performance. The hybrid framework greatly improved the diagnostic performance with 97% and 98% of classification accuracy, respectively. [2].

N. Alalwan et al. (2024), in a hybrid deep learning system to detect and classify pneumonia in chest X-ray images. The research emphasized the increasing role of CNNs in medical imaging problems and pointed out the computational burden of the processing of large-scale datasets. To overcome these drawbacks, the researchers proposed an Artificial Bee Optimization based CNN (ABO-CNN) model. Experimental tests performed on several datasets showed that the proposed method was better than some of the existing methods. The framework obtained the values of precision, accuracy, recall and F1-score as 96.95%, 88%, 86% and 86%, respectively, which makes it suitable for reliable diagnosis of pneumonia [3].

Sharma et al. (2023) The study highlighted the importance of prompt diagnosis for better outcome and mortality reduction due to pneumonia. Chest X-ray imaging is one of

the most common imaging modalities used, but manual reading is subjective and labor intensive. The researchers tackle these problems by creating a model based on VGG16 and training it with images of chest radiographs. The proposed framework successfully achieved an accuracy of 92.15%, recall score of 0.9308, F1 score 0.937, and overall classification accuracy of 0.9428, showing the effectiveness of deep learning in automated pneumonia screening [4].

Chiwariro et al. (2023) have done a comparative study of various transfer learning based deep neural network architectures to diagnose pneumonia from chest X-ray images. Five popular models, CNN, VGG16, VGG19, ResNet50, InceptionV3, and YOLOv5, were tested on the RSNA Pneumonia Detection Challenge dataset. To evaluate the model performance, validation accuracy and AUC-ROC were used. The VGG16 architecture had the validation accuracy of 88% as well as AUC-ROC of 91.75%, among the evaluated architectures. In addition, YOLOv5 showed outstanding localization performance and was able to detect pneumonia-related abnormalities with up to 99% confidence on test samples [5].

Goyal et al. (2023) presented an intelligent framework for predicting the pulmonary diseases like pneumonia and COVID-19 using chest X-ray images. The proposed system included several stages including dataset collection, image enhancement, feature extraction, and disease classification. The researchers implemented median filtering and histogram equalization for noise reduction and contrast enhancement of the images to enhance the image quality. Chest radiographs were segmented using a region growing segmentation algorithm and combined with morphological processing in order to obtain relevant regions of interest (ROI). In turn, multiple discriminatory features, such as texture, intensity, shape, and visual pattern were extracted from normalised ROI images. This study showed that using advanced preprocessing and feature engineering methods can be effective in enhancing the performance of disease detection [6].

Shankar et al. In 2023, The researchers have tested three deep learning architectures: CNN, VGG16, ResNet, and some conventional AI approach such as SVM, Naive Bayes, and KNN. The outcome showed important dissimilarity between the diagnostic accuracy of the models evaluated, highlighting the need to choose an adequate classification system. The study has revealed that the choice of the model is an important factor in the reliable, robust and overall effectiveness of automated pneumonia detection systems [7].

Table I below compares several prior studies that used deep learning and machine learning to identify pneumonia.

TABLE I. COMPARISON OF PUMONOMIA DETECTION METHODS

Authors & Year	Dataset Used	Model/Technique	Methodology Highlights	Best Performance
Opeyemi Lateef Usman et al. (2025)	Kaggle Chest X-Ray (5,856 images)	EfficientNetB1	Image normalization & augmentation; Binary Cross-Entropy loss; Adam optimizer; 25 epochs; Evaluated on 4 data splits (80:10:10, 70:15:15, 60:20:20, 75:15:10)	90.50% Accuracy, 94.97% F1-score, 0.725 ROC-AUC

Yashvi Chandola et al. (2025)	Augmented Kaggle Dataset (18,200 images)	ShuffleNet, NASNet-Mobile, EfficientNetB0 + Feature Fusion + SVM & XGBoost	Transfer learning; Deep feature extraction; Feature fusion; Hybrid ML classifiers	98% Accuracy (XGBoost), 97% (SVM); Individual CNNs: 88–90%
N. Alalwan et al. (2024)	Multiple datasets	ABO-CNN (Hybrid CNN)	Novel hybrid approach for automatic classification	96.95% Precision, 88% Accuracy, 86% Recall, 86% F1-score
Sharma, Shagun et al. (2023)	Two CXR datasets	VGG16 (Deep Learning)	Transfer learning with VGG16; Neural Network-based classification	92.15% Accuracy, F1-score: 0.937, Recall: 0.9308
Chiwariro et al. (2023)	RSNA Pneumonia Detection Challenge	VGG19, VGG16, ResNet50, InceptionV3, YOLOv5	Comparative transfer learning study	YOLOv5: 99% confidence detection; VGG16: 88% Validation Accuracy, 91.8% AUC
Goyal et al. (2023)	Two Public CXR datasets	ROI-based Feature Extraction + ML	Median filtering; Histogram equalization; Morphological ROI extraction; Texture & intensity unique characteristics	Performance metrics not clearly specified
Shankar et al. (2023)	CXR dataset	ML (KNN, DT, RF, NB, SVM) + DL (CNN, VGG16, ResNet)	Comparative evaluation of 12 models	Best model not explicitly specified; highlights importance of model selection

III. METHODOLOGY

In this part discuss the proposed efficient algorithm for pneumonia detection. First discuss the data set used for proposed research work.

A. Data Set

The Pneumonia simple's collection from Kaggle has 5,856 CXR that are divided into two primary categories: plain and Pneumonia images [22]. This dataset was specifically chosen because of its clinical reliability representation of paediatric chest X-rays, making it suitable for benchmarking deep learning-based diagnostic models. Each image in the dataset was resized to a uniform resolution of 224×224 pixels and normalized to the $[0,1]$ range to ensure computational consistency and prevent pixel-scale bias during model training. The images were converted into three-channel RGB format to align with Efficient Net's input requirements [22] [24].

For the purpose of identifying the recognition method that uses a CNN for use prediction:

B. Scaling and Normalization

It is necessary to standardize the pectoral value values of the CXR so that they fall within the series of from 0 to 1. CNNs perform better with inputs that fall within a constant range; therefore, this is significant [9].

C. Label Encoding

Encoding the labels into a numerical representation is necessary for categorization. Binary categorization usually uses the numbers zero for "Negative" and one for "Pneumonia positive."

D. Data Segmentation

The dataset was divided into three subsets to ensure effective model training, validation, and performance evaluation:

- Training Set: Use 80% of the dataset for training purposes.

- Validation Set: Datasets comprising 10% of the validation set.
- Testing Set: Ten percent of the whole data collection.

E. Data augmentation

Enhancing efficientNet B0 data for pneumonia prediction set by applying random alterations to training data augmentation might artificially increase the volume and diversity of a dataset shown in the Fig. 2. For medical imaging tasks like predicting pneumonia from chest X-rays, where:

- A significant class imbalance may exist in the dataset, with a higher number of pneumonia cases than normal cases.
- When dataset sizes are too small, over fitting might occur. To improve their performance, CNNs should learn invariant characteristics.

F. Efficient Net

A CNNs, have significantly improved automated medical image analysis. Among these architectures, EfficientNet has emerged as a powerful and computationally efficient model. EfficientNet achieves high classification accuracy while maintaining lower computational complexity compared to traditional CNN models. Transformer learning based Efficient-Net are effective at identifying patterns recognition.

Efficient Net uses compound scaling for width (w), depth (d), and resolution (r). The scaling is defined as:

$$\text{Depth} = \alpha^\phi, \text{Width} = \beta^\phi, \text{Resolution} = \gamma^\phi$$

$$\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2 \quad (1)$$

Where:

- ϕ is the compound coefficient
- $\alpha, \beta, \gamma \geq 1$ are constants

The feature extraction function is:

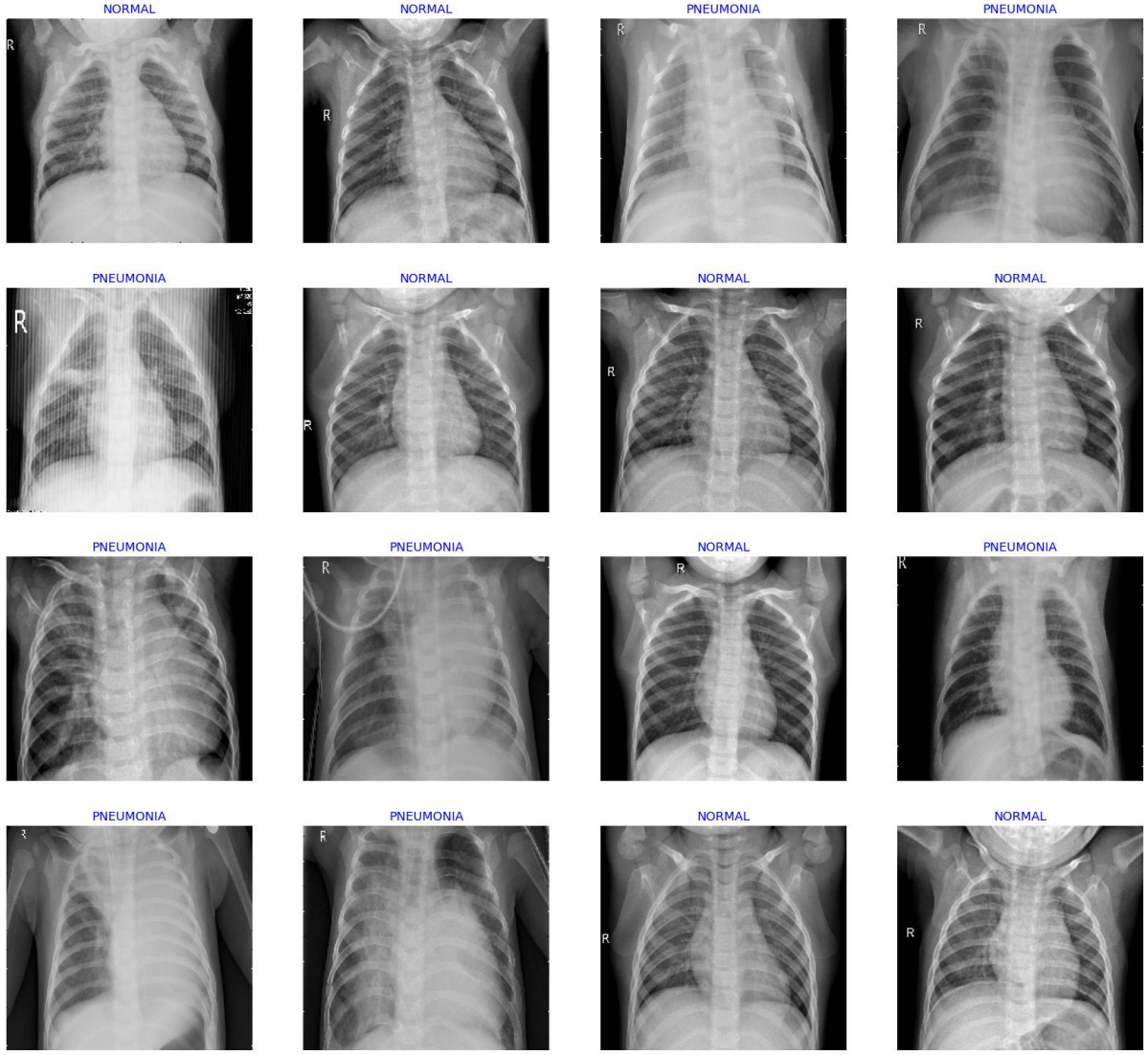


Fig. 2. Data Augmentation of Chest X- Ray images.

The feature extraction function is:

$$f_i = F(x'_i; \theta) \quad (2)$$

Where:

- F is the Efficient Net mapping function
- θ are trainable parameters
- $f_i \in \mathbb{R}^k$ is the extracted feature vector

Classification Layer

The fully connected layer computes:

$$z_i = Wf_i + b \quad (3)$$

Where:

- $w \in \mathbb{R}^{1 \times k}$
- $b \in \mathbb{R}$

Softmax activation function is used to convert the outputs of the two neurons in the final layer into class probabilities:

$$y_i = \sigma(z_i) = \frac{e^{z_i}}{\sum_{j=1}^2 e^{z_j}} \quad i = 1, 2 \quad (4)$$

Where y_i is the predicted probability of pneumonia.

TABLE II. DIFFERENT LAYER OF PROPOSED EFFICIENT NET

Layer (type)	Output Shape	Parameter
efficientnetb0	(0,1280)	4,049,571
Batch normalization	(0,1280)	5,120
dense	(0,256)	327,936
dropout	(0,256)	0
dense_1	(0,2)	514

TABLE II demonstrate the different layer and its values. Learning time-variant processes, such as medical data with dynamically altered noise circumstances, is well-suited to the capacity to modify the learning rate depending on data unique characteristics. Fig. 3 shows the CNN's architecture.

TABLE III. LOSS AND ACCURACY OUTCOME OF PROPOSED METHOD

	Loss	Accuracy
Training	0.250434547662735	0.9910714030265808

Testing	0.7609972357749939	0.968017041683197
Validation	0.291701078414917	0.9642857313156128

TABLE III shows the loss and accuracy values of training, testing and validation of proposed method. Fig. 3 illustrates an automated pneumonia detection framework that utilizes a pre-trained EfficientNet-B0 model to extract complex visual unique characteristics from preprocessed CXR images. The extracted unique characteristics are downsampled via a universal Pooling layer prior to being processed through a deep classification head comprised of three consecutive

Dense level with ReLU activations. Interleaved Dropout level (0.2 rate) are integrated between these dense steps to provide regularization and mitigate overfitting. Finally, a fully connected layer with a Softmax activation maps the network's learned representations into two distinct output probabilities: Normal or Pneumonia Detected. Fig. 4 shows the validation result outcome of proposed approach.

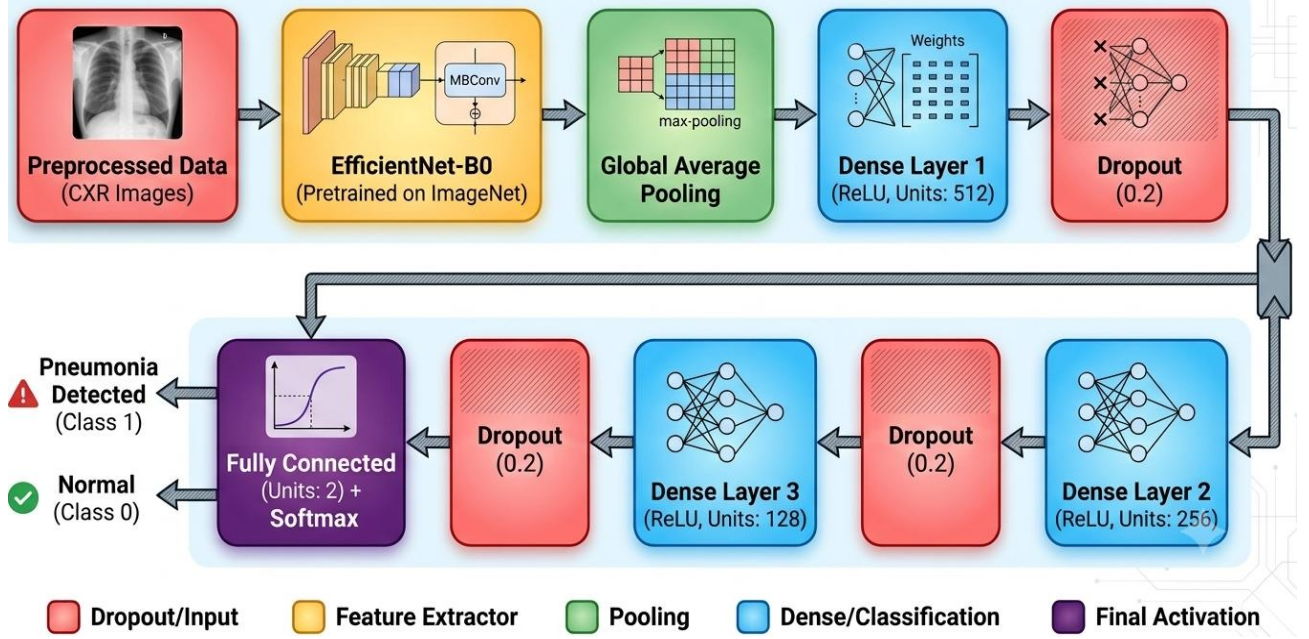


Fig. 3. Proposed Architecture of Efficient_NetB0 based pneumonia detection for X ray images

IV. RESULT AND ANALYSIS

This part will explain the analysis of the suggested approach as well as the results of the method. For the purpose of validating the suggested technique, a Python language that is based on Google Colab are required. This validation process involves optimizing a number of various outcome factors, including accuracy, precision, recall, and F-1 score, among others. For the purpose of identifying pneumonia in chest X-ray pictures, a model that makes use of CNN.

For the analysis of proposed outcome use the different parameters accuracy [8], Precision, Recall, F1-Score [10] and confusion matrix. [9].

$$Acc = \frac{TP+TN}{TP+TN+FP+FN} \quad (5)$$

$$Precision = \frac{TP}{TP+FP} \quad (6)$$

$$Recall = \frac{TP}{TP+FN} \quad (7)$$

$$F-1 \text{ Score} = \frac{2*Precision*Recall}{Precision+Recall} \quad (8)$$

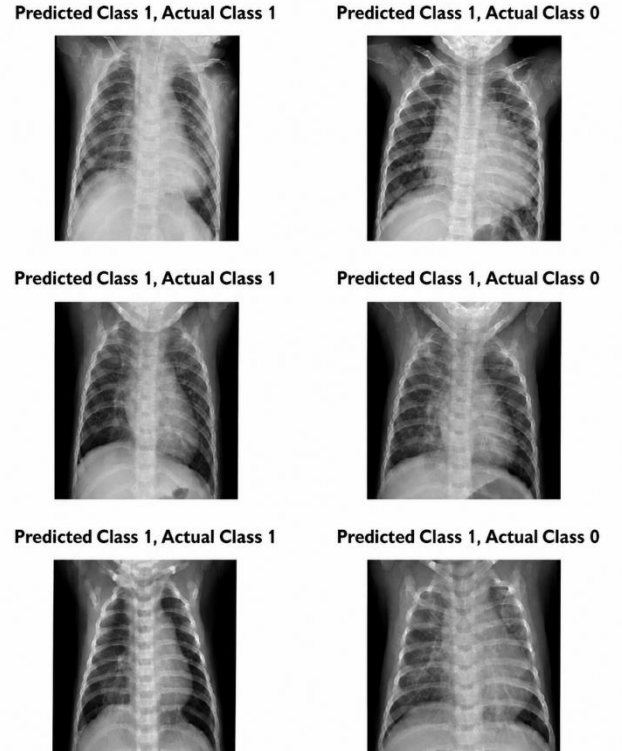


Fig. 4. Validation outcome of presented efficientNet B0

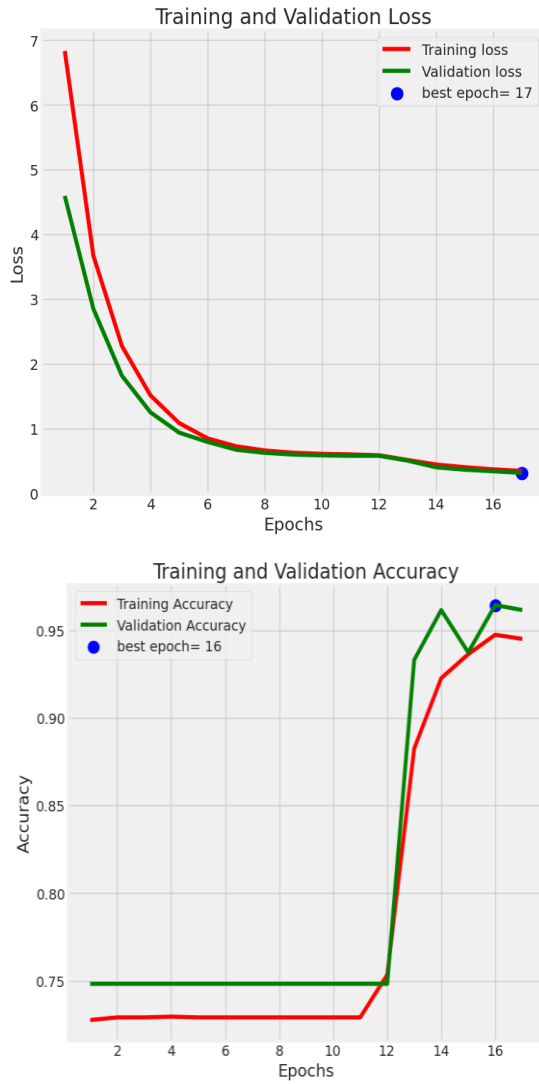


Fig. 5. Graphical representation of training and verification results. Precision and Deviation

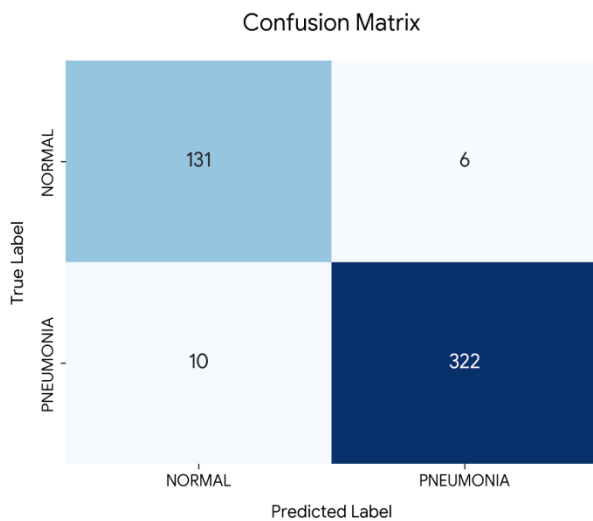


Fig. 6. Graphical representation of Confusion Matrix of presented efficientnet B0 approach

Fig. 5 exhibits the proposed method's training and validation graphics. It is possible to see the confusion matrix, which includes both TP and TN results as well as FP and FN outcomes, in Fig. 6. Confidence in real-time dataset classifications shown in Tables IV and TABLE V shows the TABLE V, Algorithm comparison between the proposed work and the previous work .

TABLE IV. PROPOSED ALGORITHM CLASSIFICATION PRECISION VALUES OUTCOME

	Precision	Recall	F1-Score	Support
Pneumonia Present	98	97	98	332
Normal Lungs	93	96	95	137

TABLE V. ALGORITHM COMPARISON BETWEEN THE PROPOSED WORK AND THE PREVIOUS WORK

Authors & Year	Dataset Used	Model / Technique	Performance compare
[1]/2025	Kaggle Chest X-Ray Dataset	EfficientNet B1	Acc. = 90.50%, F1-score = 94.97%,
[3]/2024	Multiple CXR Datasets	ABO-CNN (Hybrid CNN)	Acc. = 88%, Precision= 96.95%, Recall = 86%, F1-score = 86%
[4]/2023	Two Chest X-ray Datasets	VGG16	Acc. = 92.15%, F1-score = 93.70%, Recall =93.08%
Proposed Method/ 2026	Kaggle Chest X-ray Dataset (Normal & Pneumonia Classes)	Enhanced EfficientNet B0 + Batch Normalization + Dense Layer + Dropout	Training Acc.= 99.11% Valid. Acc. = 96.43%, Test Acc. = 96.80%, F1-score = 97%, Precision= 98%, Recall = 97%

V. SUMMARY AND FUTURE RESEARCH

This reseach presented an improved method using EfficientNet-B0 network with the aim of automatically detecting pneumonia from CXR images. The framework was able to effectively learn the visual unique characteristics of medical imaging data by pretrained deep learning. The experimental results validate that the improved EfficientNet-B0 model can be used as an extremely powerful and accurate computer-aided clinical diagnosis model. The model had an impressive training accuracy of 99.11%, validation accuracy of 96.43% and test accuracy of 96.80%. The model successfully identified 322 of the 332 real pneumonia cases, with just 10 false negatives. In healthy patients, the framework achieved a precision of 93%, and F1 score of 95%, correctly identifying 131 normal patients out of 137, while missing 6 patients with falsely classified as normal. eighted F1-score: The system achieved an overall F1-value of 97%, demonstrating stable performance in both classes despite variations in data distribution. The confusion matrix shows very few false predictions, highlighting the framework's ability to effectively reduce the risk of diagnosis..

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